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September 25, 2026
updated September 30, 2026

Statistical Models
Week 6: Algorithms, and models for count data

We will conclude the general topics about mle for now with a look at the actual algorithms behind maximum likelihood estimation as done by function `glm`: Fisher scoring and iteratively reweighted least squares.

Then we continue with particular generalized linear models, for binary and count data. For binary or binomial data (proportions of successes in Bernoulli trials), numerous link functions have been suggested. They can be motivated by a "model hidden behind the model": we imagine that the success of a Bernoulli trial depends on an unknown "latent" variable. An important application for the binomial model are contingency tables. The straightforward choice for count data are Poisson distributions. We will see that the canonical link function (log link) has a direct interpretation as "multiplicative" model.

If there is time, we will look at the theory behind the AIC criterion for model fit.

Relevant Readings

- Algorithms to find the mle: Agresti, 4.5, Davison 10.2.2 and 10.3.2.
- Models for binary and count data: Davison, 10.4, 10.5.
- Multinomial model and the "Poisson trick": Davison 10.5.2.

Lecture Material

- Link to onenotes on Google drive
- [Lecture 12]
  About the "dead flies" data set
  slides
- [Lecture 13]
  Processing and Analysis of Premier League Data
  Ship Insurance Incidences - Poisson rates

Exercises 6

Problem 6.1: LD_{50} for the "bliss" data – compare different link functions.

Dataset `bliss` from the `SMpracticals` library contains data on mortality of flour-beetles as a function of dose of the poison carbon disulphide (Bliss, 1935, Table IV). To analyze these data, Bliss used a binomial generalized linear model with probit link.

a. Fit a binomial glm with *canonical link* to the data and estimate the LD_{50}. Evaluate the fit of your model with different summary statistics and graphical methods.

b. Repeat exercise (a) with the probit link and with the complementary log-log link.

c. Would you prefer any of these three link functions? If so, why?

Problem 6.2: Model parsimonity: a simulation experiment.

(Adapted from Agresti (2015), exercises 4.36 and 4.37)
For x between 0 and 100, suppose the normal linear model holds with

$$\mathrm{E}(y) = 45 + 0.1x + 5 \cdot 10^{-4}\, x^2 + 5 \cdot 10^{-7}\, x^3 + 5 \cdot 10^{-10}\, x^4 + 5 \cdot 10^{-13}\, x^5$$

and $\sigma = 10.0$. Randomly generate $n = 25$ observations from the model, with x having a uniform distribution between 0 and 100. Fit the simple model

$$M_1: \quad \mu_j = \beta_0 + \beta_1 x_j,$$

an "intermediate" model

$$M_2: \quad \mathrm{E}(y) = \beta_0 + \beta_1 x + \beta_2 x^2,$$

and the "correct" model

$$M_5: \quad \mathrm{E}(y) = \beta_0 + \beta_1 x + \cdots + \beta_5 x^5.$$

Construct plots, showing the data, the true relationship, and the model fits. For each model, summarize the quality of the fit by the average of $|\hat{\mu}_j - \mu_j|$, that is, $\frac{1}{n}\sum_{j=1}^{n} |\hat{\mu}_j - \mu_j|$, where $\mu_j = \mathrm{E}(y_j)$ and $\hat{\mu}_j$ is the jth fitted value, depending on the model: for example in model M_1, $\mu_j = \beta_0 + \beta_1 x_j$ and $\hat{\mu}_j = \hat{\beta}_0 + \hat{\beta}_1 x_j$.
Repeat the experiment several times with different x-values. It is also interesting to look what happens if you increase the number of observations substantially, or change the true coefficients. Summarize, and explain what this exercise illustrates about model parsimony. What does the fit of the "correct" model M_5 illustrate about (multi)collinearity?

Hint

To fit polynomial models such as M_2 in R, you can directly specify the components: `y ~ I(x) + I(x^2)` or use the convenient function `poly`, namely `y ~ poly(x, 2, raw = TRUE)`.

Problem 6.3: A graphical model check.

(From Agresti, 7.28)
Other than a formal goodness-of-fit test, one analysis that provides a sense of whether a particular GLM is plausible is the following:
Suppose the ML fitted equation were the true equation. At the observed x-values for the n observations, randomly generate n variates with distributions specified by the fitted GLM. Construct scatterplots. Do they look like the scatterplots that were actually observed?
Do this for a Poisson loglinear model for the horseshoe crab data[a], with y = number of satellites and x = width. Does the variability about the fit resemble that in the actual data, including a similar number of 0's and large values? Repeat this a few times to get a better sense of how the scatterplot observed differs from what you would observe if the Poisson GLM truly held.
(Find the dataset `crabs` in the `data` folder on Google Drive)

[a] Atlantic horseshoe crabs come to the beach to mate in nights with high tide and full moon. The larger female crabs are surrounded by smaller males, called satellites, that wait to fertilize the eggs laid by the females. The `crabs.csv` file in Google drive folder `data` contains the observations for 173 female horseshoe crabs, with variables `y`: number of satellites, `weight`: weight in kg, `width`: carapax width in cm, `color`: a color category (integer valued), and `spine`: condition of the spine, an ordinal variable (integer valued)

Problem 6.4: Modelling variance in repeated experiments.

Davison, Problem 10.10.8, plus hints.

a. Show that the scaled chi-squared density with known degrees of freedom ν,

$$f(y;\sigma,\nu) = \frac{y^{\nu/2-1}}{2^{\nu/2}\sigma^{\nu}\Gamma(\nu/2)} \exp\left(-\frac{y}{2\sigma^2}\right), \quad y>0, \sigma>0, \nu = 1,2,\dots,$$

is exponential dispersion family. Express the canonical parameter θ and the dispersion parameter ϕ as functions of ν and σ^2.
Hence derive an expression for its r th cumulant, $r \geq 1$.

Hint

- Consider ν as a known constant. Several choices for θ and ϕ are possible, and ϕ is a constant, too. One possible choice is $\phi = -2$; with this choice, the canonical link function depends on ν. Another option is $\phi = -2/\nu$ — check it out :-) If you need a reminder for the moment generating function look at the top of Davison, page 481. Then you easily recall the cumulant generating function.
- To control your calculations: the second cumulant is $\kappa_2 = 2\nu\sigma^4$, the variance of y.

b. The yield of an industrial process was measured r_i times independently at m different temperatures t_i . The resulting yields $Z_{ij}, i = 1,\dots,m, j = 1,\dots,r_i$ may be assumed to be independent and normally distributed with both means ζ_i and variances τ_i dependent on t_i.

Explain how the sums of squares $Y_i = \sum_{j=1}^{r_i}(Z_{ij} - \overline{Z}_i)^2$, where $\overline{Z}_i = \frac{1}{r_i}\sum_{j=1}^{r_i} Z_{ij}$, may be used to assess the dependence of variance on temperature in a suitable generalized linear model.

Briefly discuss the advantages and disadvantages of the canonical link function of your model.

Problem 6.5: Deviance for the binary logistic model.

Davison, Exercises 10.4.1.
This exercise deals with data $y_1,\dots,y_n$ that come frome independent Bernoulli experiments with probability parameter $\pi_1,\dots,\pi_n$.
A little extra before solving parts a-c: identify the distribution of Y_j with $\mathrm{P}(Y_j = 1) = \pi_j$ and $\mathrm{P}(Y_j = 0) = 1 - \pi_j$ as an exponential dispersion distribution with dispersion parameter $\phi = 1$. Determine the canonical parameter θ and the function b.
Here is the exercise text from Davison:
Data $y_1,\dots,y_n$ are assumed to follow a binary logistic model in which y_j takes value 1 with probability $\pi_j = \exp(x_j^{\mathsf{T}}\beta)/(1+\exp(x_j^{\mathsf{T}}\beta))$ and value 0 otherwise, for $j = 1,\dots,n$.

a. Show that the deviance for a model with fitted probabilities $\hat{\pi}_j$ can be written as

$$D = -2\left(y^{\mathsf{T}}X\hat{\beta} + \sum_{j=1}^{n}\log(1-\hat{\pi}_j)\right)$$

and that the likelihood equation is $X^{\mathsf{T}}y = X^{\mathsf{T}}\hat{\pi}$. Hence show that the deviance is a function of the $\hat{\pi}_j$ alone.

b. If $\pi_1 = \cdots = \pi_n = \pi$, then show that $\hat{\pi} = \bar{y}$, and verify that

$$D = -2n\left(\bar{y}\log\bar{y} + (1-\bar{y})\log(1-\bar{y})\right).$$

Comment on the implications for using D to measure the discrepancy between the data and fitted model.

c. Assuming that $\pi_1 = \cdots = \pi_n = \pi$ (as in part (b)), show that Pearson's statistic, formula (10.21) in Davison,

$$P = \frac{1}{\phi} \sum_{j=1}^{n} \frac{(y_j - \hat{\mu}_j)^2}{V(\hat{\mu}_j)/a_j},$$

is identically equal to n. Comment.

Hints

for (a), observe that the likelihood of the saturated model is $L(\pi) = 1$ (this is also the likelihood for the parameter β of the glm).
In (c): Rmember the definition of a_j in this context: $a_j = \phi_j/\phi$. In this exercise, use $\phi = 1$ and $a_j = 1$ in formula (10.21).